

INTRODUCTION to BLOCKCHAIN

CHAPTER3: CRYPTOCURRENCIES

Dr. Noureddine Lasla

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Chapter Overview

Objective:

Understanding Digital Assets and Their Role in Blockchain

Key Areas of Focus:

- Introduction to Money
- History of Money
- Introduction to Cryptocurrencies
- Types of Cryptocurrencies
- Key Features of Cryptocurrencies
- Bitcoin
- Altcoin

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What is Money?



Money is any item or medium of exchange that symbolize perceived **VALUE**.

Functions of Money:

- Medium of exchange: Money eliminates the inefficiencies of **barter** systems by providing a commonly accepted method for exchanging goods and services.
- Unit of account: Money provides a standard measure of value, allowing people to compare the worth of different goods and services. (**Abstract units of measure.)**
- Store of value: Unlike perishable goods, which may deteriorate over time, money retains its value.
- Standard of Deferred Payment: It provides a mechanism for credit and financial contracts, making it possible to lend, borrow, and repay in a universally accepted medium.

History of money



Barter

trade depends on a "**double coincidence of wants**,"

Commodity Money

Early forms of money included items with **intrinsic value** (salt, livestock, shell)

Metal Coins

400 BCE: standardized, and their value was backed by the weight and type of metal



Paper Money

700 AD: represent a **promise to exchange** for metal coins or goods. To avoid carrying large amount of coins

Banking & Bills of Exchange

Middle Age: Method of settling accounts in international trade



Gold Standard

!9th Century: Money was directly **linked** to a specific quantity of **gold**

Fiat Money

20th Century: not backed by a physical commodity. Value is derived from the **trust to Gov.**

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History of money

Monetary System	Examples	Backing	Main Feature
Fiat Money	U.S. Dollar, Euro, Yen	No backing (trust-based)	Government-issued currency
Commodity Money	Gold Standard (historical)	Gold, Silver	Backed by physical commodities
Representative Money	Gold certificates (historical)	Physical commodity (e.g., gold)	Claims to a commodity
Digital/Cryptocurrencies	Bitcoin, Ethereum	Decentralized (peer-to-peer)	Digital and decentralized
CBDCs (Central Bank Digital Currencies)	Digital Yuan, Sand Dollar	Government-issued fiat currency	Digital fiat currency, central bank control

Again…, what is money?

Units of currency are merely **abstract unit** of measurements.

Money is used to measure debt: It is an IOU (I owe you)

Technically any one can create money. But the hard part is to get that money **accepted**.

Introduction to Cryptocurrencies

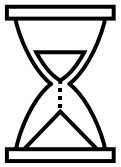
What Are Cryptocurrencies?

- Digital or virtual currencies secured by cryptography.
- Operate on decentralized networks (blockchain).
- Enable peer-to-peer transactions without intermediaries (**electronic cash**).



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History of Cryptocurrencies



- 2008: Bitcoin whitepaper published by Satoshi Nakamoto.
- 2009: Bitcoin network launched; first block mined (Genesis Block) *03-01-2009*.
- 2011: Emergence of altcoins (e.g., Litecoin).
- 2015: Ethereum introduced smart contracts.
- 2017: ICO (Initial Coin Offering) boom.
  - **Filecoin ICO (2017)** raised $205 million
- 2020s: Rise of DeFi, NFTs, and institutional adoption.





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Types of Cryptocurrencies

- **Bitcoin** (BTC): First cryptocurrency, store of value, Limited supply (21 million coins).



- **Altcoins**: Ethereum (ETH), Litecoin (LTC), Ripple (XRP), etc.
  - Reduce volatility and act as a bridge between crypto and traditional finance.



- **Stablecoins**: Pegged to fiat currencies (e.g., USDT, USDC).



- **Tokens**: Used within specific ecosystems (e.g., Binance Coin, linkchain, etc.).

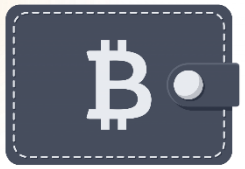
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Bitcoin

- Bitcoin transactions are transferred directly between users through a **peer-to-peer** network.
- Each transaction is recorded on the blockchain, ensuring that all Bitcoin exchanges are secure, transparent, and immutable.
- Bitcoin transactions rely on **asymmetric cryptography** for authentication and security.

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Bitcoin Wallet



Digital tool that allows users to **store, send, and receive Bitcoin**.

- Private key: **proves ownership** of Bitcoins.
- Public key: **address** linked to the private key, used to receive Bitcoin.
- Types of Bitcoin Wallets:
  - Hot Wallets (online, mobile, desktop wallets): connected to the internet, for **quick** transactions.
  - Cold Wallets (offline, Hardware, paper wallets): providing enhanced security for **long-term** storage of Bitcoin.

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Bitcoin UTXO

Bitcoin uses the **UTXO** (Unspent Transaction Output) model instead of a **balance-based** system.

In Bitcoin, every transaction consumes inputs (previously unspent outputs) and creates outputs (new UTXOs). These outputs can then be spent in future transactions.

Rather than maintaining a running balance for each address, Bitcoin tracks all individual UTXOs, each representing a specific amount of Bitcoin.

Bitcoin UTXO

UTXO (Unspent Transaction Output)

UTXO represents the amount of Bitcoin that has not been spent from a transaction.

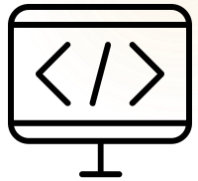
- Each Bitcoin transaction involves spending previously received UTXOs.
- Transaction **Inputs**: When you send Bitcoin, you are using one or more of your previous UTXOs as inputs.
- Transaction **Outputs**: The outputs are the new UTXOs created by the transaction. Any leftover Bitcoin after the transaction (i.e., **change**) is also returned as a new UTXO.

Bitcoin UTXO

How UTXOs Work:

- When you receive Bitcoin, you are essentially receiving UTXOs associated with a specific amount.
- When you make a transaction, you use these UTXOs as inputs. Bitcoin's system **doesn't allow** spending fractions of a UTXO, so if you're sending less than the full amount, the **remainder is sent back to you** as a new UTXO (often called "change").
- A transaction can have **multiple inputs** (coming from different UTXOs) and **multiple outputs** (sending Bitcoin to one or more recipients).

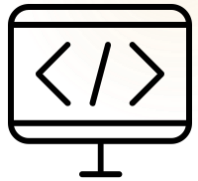
Bitcoin Scripting Language



What is Bitcoin script?

- Bitcoin Script is a **stack-based** programming language used to define the rules for how Bitcoin transactions are processed.
- It is **not** Turing-complete, meaning it's not designed for complex computations, but rather for simple transaction validation and authorization.
- Work with two types of instructions: **data instructions** and **OP_CODE**.

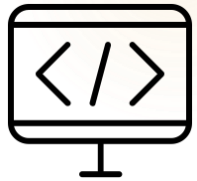
Bitcoin Scripting Language



Purpose of Bitcoin Script

- The goal is to define the conditions under which a Bitcoin transaction can be spent.
- It's a language for locking and unlocking Bitcoin during a transaction.
  - Lock script (**scriptPubKey**)
  - Unlock script (**scriptSig**).

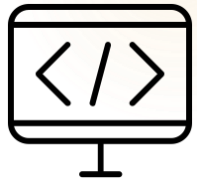
Bitcoin Scripting Language



Features of Bitcoin Script

- Stack-Based: uses a stack data structure where operations are performed by **pushing** and **popping** data to and from the **stack**.
- Non-Turing Complete: is **limited** in functionality to **prevent** complex operations that could cause security issues or make the system inefficient.
- Simple Operations: supports **basic** operations like addition, subtraction, comparison, and logical AND/OR operations.

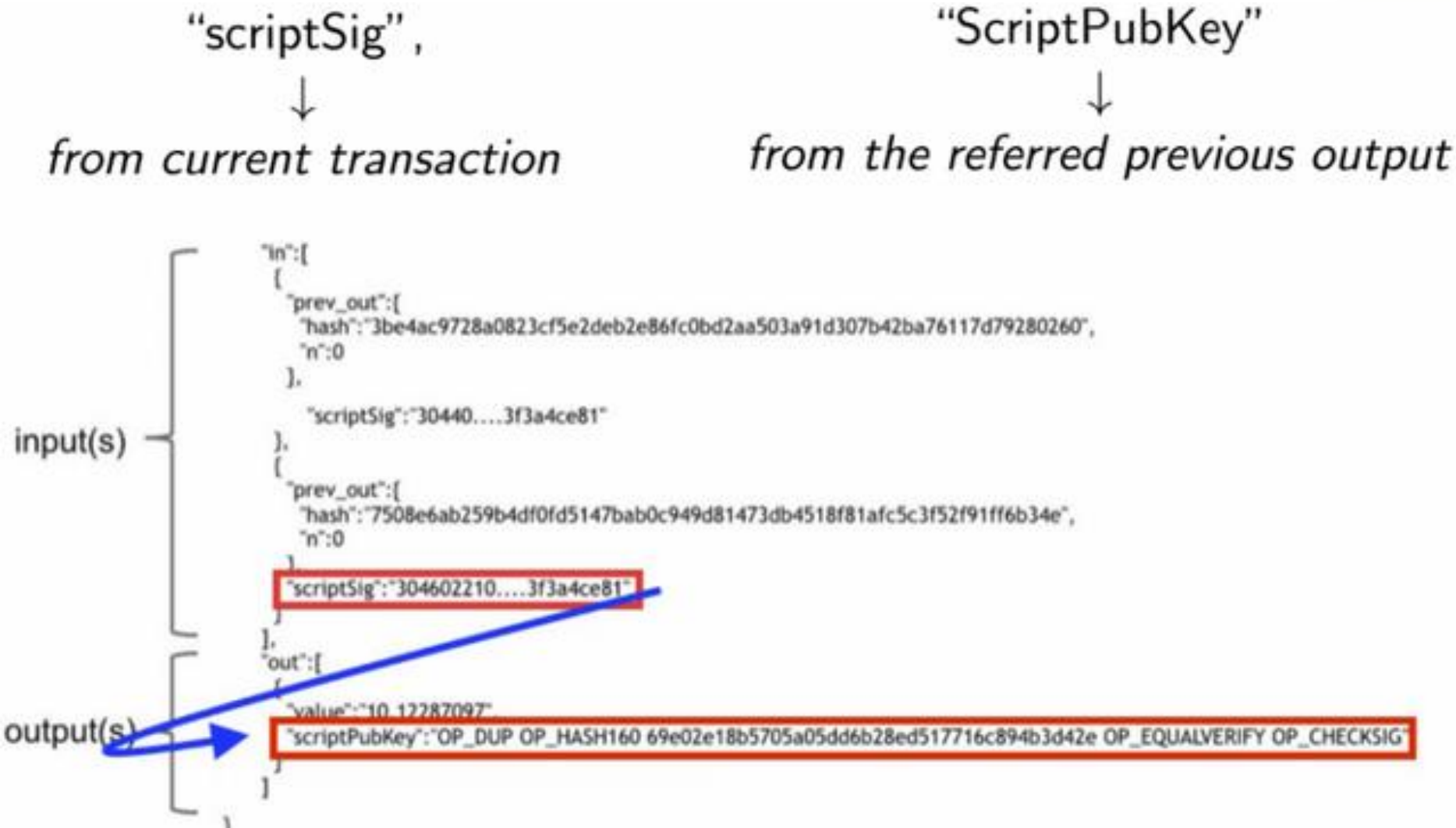
Bitcoin Scripting Language



How Bitcoin Script Work

- Locking Bitcoin:
  - When a Bitcoin is sent, the sender specifies conditions under which it can be spent by creating a **scriptPubKey** (a locking script).
- Unlocking Bitcoin:
  - When the recipient wants to spend the Bitcoin, they must provide an unlocking script (**scriptSig**) to satisfy the conditions in the **scriptPubKey**.

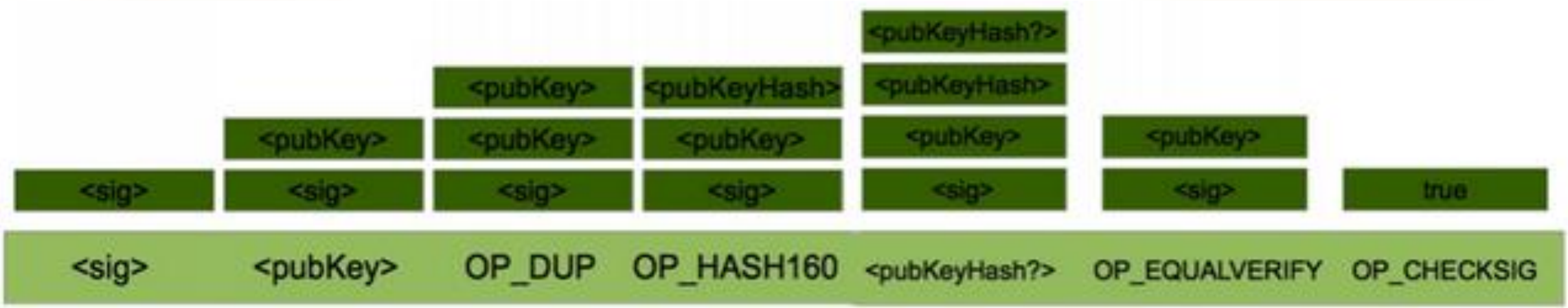
Bitcoin Scripting Language



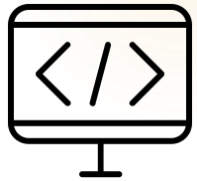
Bitcoin Script Language



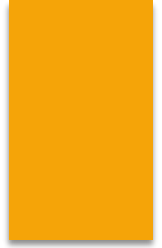
Bitcoin Script Language



Bitcoin Script Language



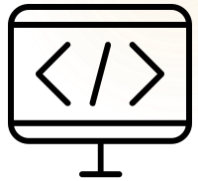




Summary of the different types of Bitcoin scripts

Type of Script	Description	Example
Pay-to-PubKey-Hash (P2PKH)	The most common script type. The funds are locked to a public key hash (Bitcoin address). Only the owner of the private key corresponding to the public key hash can spend the funds.	**scriptPubKey**: OP_DUP OP_HASH160 <public_key_hash> OP_EQUALVERIFY OP_CHECKSIG
Pay-to-Script-Hash (P2SH)	The payer sends funds to a script hash, which can then be spent using any valid script matching the hash. Commonly used for multi-signature addresses and more complex scripts.	**scriptPubKey**: OP_HASH160 <script_hash> OP_EQUAL
Multi-Signature (Multisig)	A script type where multiple signatures are required to unlock the funds. Allows the creation of joint wallets, where several participants must sign to spend funds.	**scriptPubKey**: OP_2 <pubKey1> <pubKey2> <pubKey3> OP_3 OP_CHECKMULTISIG
Pay-to-Witness-PubKey-Hash (P2WPKH)	A SegWit-based script where the funds are locked to a public key hash and spent using a **witness**. It reduces transaction size and improves efficiency.	**scriptPubKey**: 0 <public_key_hash>
Pay-to-Witness-Script-Hash (P2WSH)	A SegWit-based script that locks funds to a script hash. Similar to **P2SH**, but designed to take advantage of SegWit.	**scriptPubKey**: 0 <script_hash>

Bitcoin Script Language





Summary of Bitcoin Script opcodes

Opcode	Description
OP_DUP	Duplicates the top item on the stack.
OP_HASH160	Applies **SHA-256** followed by **RIPEMD-160** hashing to the top item.
OP_CHECKSIG	Verifies that the digital signature is valid for the provided public key and data.
OP_EQUAL	Checks if the top two items on the stack are equal.
OP_EQUALVERIFY	Like **OP_EQUAL**, but also removes the top two items if they are equal.
OP_CHECKMULTISIG	Verifies multiple signatures for a **multi-signature** transaction.
OP_RETURN	Marks a transaction output as **unspendable**. Used for embedding data in the blockchain.
OP_IF / OP_ELSE	Conditional logic: executes script based on the truth value of the preceding condition.
OP_VERIFY	Verifies that the top stack item is **true** (non-zero). If not, the script fails.
OP_NOP	Does nothing, often used for backward compatibility or future extensions.

Bitcoin TX Example

How to transfer some BTC between Alice, Bob and Joe?

TxID:42b6a77cf6096f52fc7513cfdf861bcc1e841d7b05eb48d095a764df79501c7d

INPUTS From			OUTPUTS To	
(from previous transaction Alice has received):				
Alice	0.1005 BTC	⇒	Output #0 Bob's address	0.1000 BTC (spent)
			Transaction fees	0.0005 BTC

1. Bob's address (receiver): hash(Bob's public key)
2. Proof of ownership: transaction signed by Alice's private key (unlock)
3. Propagate the transaction on the Bitcoin network
4. Validate the transaction (rules):
   - check syntax, size
   - check double spending
5. Mining: the transaction becomes part of the blockchain

Bitcoin TX Example



Transfer from Bob to Joe

TxID: 30076701180f4dd48a7f1bb37027c6e791e950eaa44b6449f6b5d660b523f967

INPUTS From			OUTPUTS To	
42b6a77cf6096f52fc7513cfdf861bcc1e841d7b05eb48d095a764df79501c7d:				
Bob	0.1000 BTC	⇒	Output #0 Joe's address	0.0150 BTC (spent)
			Output#1 Bob's address (change)	0.0845 BTC (unspent)
			Transaction fees	0.0005 BTC

Bitcoin uses a scripting language (Turing incomplete)

The script for a Bitcoin transfer to dest. Bitcoin address D simply encumbers future spending of the bitcoins with two things:

- the spender must provide a public key that, when hashed, yields destination address D embedded in the script, and
- a signature to prove ownership of the private key corresponding to the public key just provided.

Altcoins



Altcoins are any cryptocurrencies that are not Bitcoin. The term "Altcoin" is short for "**alt**ernative **coin**," meaning they are alternatives to Bitcoin.

- Altcoins offer a wide range of applications and use cases.
  - Smart contracts (Ethereum), DeFi (Uniswap, Aave)
  - Privacy-focused transactions (e.g., Monero, Zcash)
  - Stable value (e.g., Tether, USDC)
- Many altcoins are at the forefront of blockchain innovation, exploring new consensus algorithms, scalability solutions, and other features.

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Altcoins



Popular Altcoins

Altcoin	Use Case	Blockchain
Ethereum (ETH)	Smart contracts, dApps	Ethereum
Binance Coin (BNB)	Exchange token, DeFi	Binance Chain
Cardano (ADA)	Scalable smart contracts	Cardano
Solana (SOL)	High-speed transactions	Solana
Polkadot (DOT)	Blockchain interoperability	Polkadot
Dogecoin (DOGE)	Meme coin, payment method	Dogecoin

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